26900235 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEFSQ305 - PROCESSING TECHNOLOGY OF FRUITS AND VEGETABLES

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Rhizomes are modified underground stems that generally grow [K] / [CO1]

   a). Vertically upward b). Horizontally

   c). Vertically downward d). Above the soil surface

2. Recall the major cause of drip loss on thawing. [U] / [CO3]

   a). Ice-crystal damage to cell walls and membranes b). Low sugar concentration

   c). Moisture-proof packaging d). Stable frozen storage

3. Dehydration of fruits and vegetables mainly aims at reducing the _____. [K] / [CO3]

   a). Protein b). Moisture c). Fat d). Fibre

4. Identify the main ingredient used in the preparation of jam. [K] / [CO4]

a). Fruit pulp, sugar and pectin

b). Fruit juice, water, starch

c). Fruit juice, cellulose, sugar

d). Fruit juice, milk, sugar

5. The outermost layer of the pericarp [K] / [CO1]

a). Exocarp b). Mesocarp

c). Endocarp d). Epidermis

6. Identify the main preservation principle involved in canning [K] / [CO3]

a). Freezing

b). Heat processing and hermetic sealing

c). Fermentation

d). Drying

7. Identify the main purpose of blanching tomatoes before puree preparation. [U] / [CO4]

a). Increase acidity b). Facilitate skin removal

c). Increaase salt content d). Reduce colour

8. Betalains are characteristic pigments of [K] / [CO1]

a). Beetroot b). Tomato

c). Spinach d). Carrot

9. A food processor needs to reduce surface microbial contamination on freshly harvested lettuce. Indicate the most suitable treatment: [A] / [CO3]

a). Washing with plain water only

b). Treatment with electrolyzed water

c). Dry heating at high temperature

d). Freezing before washing

10. Select the reason for fermentation occur in salt pickles [U] / [CO4]

a). Beneficial microorganisms b). Insects

c). Minerals d). Fat

11. Select the major purpose of ozone treatment in fruit processing. [U] / [CO3]

a). Increasing sugar content

b). Microbial reduction and surface sanitation

c). Increasing moisture content

d). Increasing ripening

12. Select from the options,The alcoholic strength of cider is generally lower than of [U] / [CO4]

a). Fruit juice b). Beer and wine

c). Squash d). Nectar

13. Identify the pretreatment used to inactivate peroxidase and lipoxygenase before freezing vegetables. [U] / [CO3]

a). Grading

b). Blanching

c). Thawing

d). Packaging

14. Indicate the concentration of tomato paste is increased by [U] / [CO4]

a). Evaporation of water b). Addition of water

c). Fermentation d). Carbonation

15. In top icing, fruits and vegetables are cooled using [K] / [CO1]

a). Ice cubes b). Ice slurry

c). Crushed ice d). All of these

16. Select the quality change that can be delayed by applying a suitable edible coating to fresh fruits. [U] / [CO3]

a). Moisture loss

b). Protein synthesis

c). Mineral formation

d). Fibre production

17. Indicate the componenet present in high proportion in syrup [U] / [CO4]

a). Salt b). Sugar

c). Protien d). Fat

18. Recognise the process responsible for water removal during freeze drying. [U] / [CO3]

a). Evaporation

b). Condensation

c). Melting

d). Sublimation

19. The enzyme mainly responsible for enzymatic browning in fruits [K] / [CO1]

a). Polygalacturonase b). Pectin methyl esterase

c). Protopectinase d). Polyphenol oxidase

20. Identify the preservation method involving removal of water from fruits. [U] / [CO4]

a). Freezing b). Canning

c). Dehydration d). Fermentation

[1x20 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

21. Write a note on factors affecting the jelly preparation. [A] / [CO4]

22. List the major pigments present in fruits and vegetables and state their characteristic colours. [K] / [CO2]

23. Explain simultaneous heat transfer and mass transfer during drying. [A] / [CO3]

24. Explain fermented beverage and classify them with suitable examples [U] / [CO4]

25. Explain the different methods used in cleaning of fruits and vegetables. [U] / [CO1]

26. Describe the physiological, flavour, taste, colour, texture and nutritional changes occurring during refrigerated storage of fruits and vegetables. [U] / [CO2]

27. Explain the processing of wine with a flow chart [U] / [CO4]

28. What are food preservatives? Explain their importance in fruit processing. [U] / [CO3]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

29. Discuss the principles and applications of controlled atmosphere packaging and modified atmosphere packaging in fresh produce storage. [U] / [CO2]

30. Classify the major freezing methods based on mode of heat transfer and present suitable applications for each method. [A] / [CO3]

31. Explain the methods used to control post-harvest losses. [U] / [CO1]

32. Discusss the factors affecting vinegar production, including temperature,acidity,oxygen supply and microbial activity [U] / [CO4]

[10x2 = 20]